

Multidimensional routine hematologic modeling for acute myocardial infarction phenotype discrimination in coronary artery disease: a single-center retrospective study

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Abstract

Background: Acute myocardial infarction (AMI) is accompanied by inflammatory, platelet, coagulation, and erythrocyte responses. Routine complete blood count (CBC) measurements capture several of these changes but are often reduced to fixed ratios. We examined whether joint modeling of routine hematologic measurements could improve AMI phenotype discrimination beyond the pan-immune-inflammation value (PIV) in patients with coronary artery disease.

Methods: This single-center retrospective study included 1,820 patients treated at the First Affiliated Hospital of Xinjiang Medical University from 1 January 2020 to 1 January 2026; 453 met the definite AMI definition. Core-7 combined neutrophil, lymphocyte,

monocyte, platelet, mean platelet volume, red cell distribution width, and hemoglobin using elastic-net logistic regression. Performance was assessed by repeated five-fold cross-validation with preprocessing and tuning restricted to training data. PIV was the principal comparator, the Enhanced model added fibrinogen, and temporal sensitivity analysis used a six-month buffer based on the deidentified hospitalization/coronary-angiography date axis.

Results: PIV showed the highest discrimination among conventional inflammatory indices (area under the receiver operating characteristic curve [AUC] 0.708, 95% confidence interval [CI] 0.678-0.735). Core-7 achieved an AUC of 0.725 (95% CI 0.697-0.753), a modest improvement over PIV (Δ AUC 0.017, 95% CI 0.003-0.031). Enhanced achieved an AUC of 0.735 (95% CI 0.710-0.763) and improved on Core-7 by 0.010 (95% CI 0.003-0.017). In the later group of the temporal sensitivity analysis, Core-7 and PIV had similar AUCs (0.701 and 0.699; Δ AUC 0.002, 95% CI -0.021 to 0.025), and Enhanced had an AUC of 0.719; this analysis was not formal temporal validation.

Conclusions: Joint modeling of routine hematologic measurements provided modest additional information beyond a fixed inflammatory ratio in internal evaluation. Core-7 retained moderate discrimination in the temporal sensitivity analysis, although its incremental advantage over PIV attenuated. Fibrinogen provided a small additional increment over Core-7 in internal validation. Further independent, time-anchored evaluation is warranted; these findings do not establish diagnostic or triage readiness.

Keywords: acute myocardial infarction; coronary artery disease; complete blood count; hematologic biomarkers; fibrinogen; internal validation; calibration

Background

Acute myocardial infarction (AMI) in patients with coronary artery disease (CAD) reflects the interaction of plaque inflammation, thrombosis, platelet activation, myocardial injury, and

systemic stress [1,2]. These processes are accompanied by coordinated changes in circulating blood cells and coagulation proteins. Neutrophilia and lymphocyte depletion may accompany acute inflammatory and neurohumoral stress responses, monocytes participate in plaque inflammation and subsequent immune activation, and platelet turnover and activation can alter both platelet count and mean platelet volume (MPV) [1,3-14]. Erythrocyte indices such as red cell distribution width (RDW) and hemoglobin may reflect erythropoietic stress, systemic inflammation, comorbidity, and oxygen-delivery reserve, while fibrinogen links the acute-phase response with thrombosis [15-17]. Because these complete blood count (CBC) and coagulation measurements are obtained routinely in cardiovascular care, they provide a clinically accessible window into several biological processes relevant to AMI.

Previous studies have mainly evaluated hematologic inflammation through derived indices such as the neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), systemic immune-inflammation index (SII), systemic inflammation response index (SIRI), and pan-immune-inflammation value (PIV) [3-12]. These indices are inexpensive and easy to calculate, but each applies a fixed mathematical weighting to a limited set of cell counts. Other routinely available measurements, including MPV, RDW, and hemoglobin, are therefore often considered separately even though they may provide complementary information about the acute coronary state [13-16].

We hypothesized that evaluating these measurements together would better represent the multidimensional hematologic response accompanying AMI than a single predefined ratio. The Core-7 model was therefore built around three clinically recognizable components: leukocyte-related inflammatory responses (neutrophils, lymphocytes, and monocytes), platelet-related features (platelet count and MPV), and erythrocyte-related features (RDW and hemoglobin). Fibrinogen was examined separately as an additional coagulation and acute-phase marker [13-17]. This biologically informed grouping provided the rationale for joint modeling rather than simply selecting variables from the available laboratory database.

Accordingly, we developed and internally evaluated a multidimensional routine hematologic model for discrimination of an AMI phenotype among patients with coronary artery disease. We compared the CBC-only Core-7 model with established inflammatory indices, particularly PIV, and examined whether the addition of fibrinogen provided further information. Prespecified diagnostic and temporal sensitivity analyses were used to assess the robustness of the observed associations.

Methods

Study design and setting

This was a single-center retrospective observational study conducted in the Department of Cardiology, The First Affiliated Hospital of Xinjiang Medical University, Urumqi, Xinjiang, China, using records from Coronary Heart Disease Unit I. The study period was 1 January 2020 to 1 January 2026, and the source population comprised patients who underwent coronary angiography during this period. Because a complete eligible-patient screening log was unavailable, consecutive enrollment could not be confirmed. For the temporal sensitivity analysis, institutionally deidentified hospitalization/coronary-angiography dates were shifted on a patient-specific basis within +/-182 days. The study evaluated retrospective AMI phenotype discrimination rather than future-event prediction.

Eligibility and study population

The source export contained 2,548 records representing 2,279 unique patients. Of these, 2,010 had one source record and 269 had two; 269 surplus records were removed when retaining one patient-level record by a completeness-based rule. The source did not establish that multiple records represented distinct admissions. Among the 2,279 patients, 1,944 had discharge-diagnosis text: 456 met the definite AMI definition, 1,372 the definite non-AMI CAD definition, five had an ambiguous MI mention, and 111 did not meet the CAD/AMI target. The

remaining 335 patients had no discharge diagnosis. Before applying the core CBC availability requirement, 1,828 patients had a definite AMI or non-AMI CAD phenotype.

Outcome definition

AMI phenotype was classified from discharge-diagnosis text; no independent clinical adjudication was available. Definite AMI required explicit current or acute/subacute MI language, including unambiguous ST-segment elevation myocardial infarction (STEMI), non-ST-segment elevation myocardial infarction (NSTEMI), or an acute wall-specific MI description. Old or prior MI, MI history, prior percutaneous coronary intervention (PCI), CAD, stable angina, and unstable angina were not sufficient alone. An MI keyword without clear acute or old/prior context, missing diagnosis, or uncertainty markers was retained as ambiguous/review. A definite non-AMI CAD phenotype included CAD, coronary disease, angina, or revascularization descriptions without current acute MI, with old/prior MI allowed in the broad primary control but excluded from the strict control. Troponin, creatine kinase-MB (CK-MB), and other cardiac injury markers were supportive only; incomplete timing, reference-limit, and dynamic-change information prevented complete Fourth Universal Definition adjudication [2].

Primary and strict-control cohorts

The cohort-entry rule required non-missing absolute neutrophil, lymphocyte, monocyte, and platelet measurements. Eight patients failed this four-variable requirement: 3 with AMI and 5 with non-AMI CAD. Among the resulting primary cohort, all seven Core-7 predictors (neutrophils, lymphocytes, monocytes, platelets, MPV, RDW, and hemoglobin) were complete, so the Core-7 complete-case analysis retained all N=1820 records. The strict-control sensitivity cohort included all definite AMI records meeting the CBC requirement and only stable angina, unstable angina, or CAD-only controls without current MI, old/prior MI, or an ACS-only marker; it contained N=1517 patients, including 453 AMI and 1064 controls.

Predictors and laboratory measurements

The Core-7 predictors were absolute neutrophil, lymphocyte, and monocyte counts, platelet count, MPV, RDW, and hemoglobin; fibrinogen was added for the Enhanced model. The hematologic measurements used for analysis were obtained in association with the target hospitalization. For patients with multiple source records, one record was selected in descending priority: non-empty discharge-diagnosis information; greater completeness of absolute neutrophil, lymphocyte, monocyte, and platelet values (0-4); presence of a parsable CBC/WBC timestamp; higher qc_nonmissing_count; and original source-row order as the final tie-break. Historical timestamp fields in the exported research dataset contained legacy registration inconsistencies and could not reliably reconstruct exact CBC sampling time relative to admission, coronary angiography, or AMI diagnosis for every patient. Accordingly, the selection rule was not a clinical-time hierarchy, and the measurements are not described as uniformly first-admission, pre-angiography, pre-diagnostic, or pre-treatment. Laboratory units were $10^9/L$ for leukocyte and platelet counts, fL for MPV, percent for RDW, and g/L for hemoglobin and fibrinogen. Analyzer, reagent, and platform details were not included in the deidentified research database but can be retrieved from institutional laboratory records if required during peer review. Internal CBC consistency was checked by comparing recorded absolute differential counts with values calculated from total white-cell counts and differential percentages; absolute counts were used for all derived inflammatory indices.

Conventional inflammatory indices

Using absolute counts, we calculated $NLR = \text{Neut}/\text{Lymph}$, $PLR = \text{PLT}/\text{Lymph}$, the monocyte-to-lymphocyte ratio ($MLR = \text{Mono}/\text{Lymph}$), $SII = \text{PLT} * \text{Neut}/\text{Lymph}$, $SIRI = \text{Neut} * \text{Mono}/\text{Lymph}$, $PIV = \text{PLT} * \text{Neut} * \text{Mono}/\text{Lymph}$, and the hemoglobin-to-RDW ratio ($HRR = \text{Hb}/\text{RDW}$ [3-17]). PIV was the prespecified primary comparator; $\log(1 + PIV)$, training-fold standardization, and an unpenalized logistic model were used on the same patients and outer-fold partitions as Core-7.

The other indices were reported as descriptive benchmarks; no additional pairwise Core-7-versus-index tests were performed. All indices were assessed within the same repeated nested cross-validation framework and remained continuous without data-derived cutoffs.

Model development and preprocessing

Core-7 used the seven CBC predictors, and the Enhanced model additionally included fibrinogen. Elastic-net logistic regression was evaluated across alpha values of 0.25, 0.50, and 0.75 and C values of 0.10, 1.00, and 10.00, with parameter selection performed by five-fold cross-validation within the training data [18,19]. Neutrophils, lymphocytes, monocytes, platelets, and fibrinogen were log1p transformed and standardized; MPV, RDW, and hemoglobin were standardized on their original scale. Core-7 used complete cases. For the Enhanced model, missing fibrinogen values were replaced using the median estimated from the corresponding training data. Preprocessing, imputation, transformation, standardization, and parameter selection were performed separately within each training fold to prevent information leakage. To assess whether hematologic information added to basic clinical characteristics, the prespecified baseline clinical model included age, sex, hypertension, and diabetes; smoking was not available. Hypertension was missing in 79/1,820 patients (4.34%) and diabetes in 48/1,820 (2.64%); both were imputed using medians estimated within each training fold, while age and sex were complete. Clinical+PIV, Clinical+Core-7, and Clinical+Enhanced were evaluated using the same cross-validation procedure.

Internal validation and uncertainty

Primary internal validation used stratified outer five-fold cross-validation repeated 10 times, with five-fold hyperparameter tuning within each outer training set. Each patient received one held-out probability per repeat; the arithmetic mean of the 10 held-out probabilities was used for patient-level evaluation. Identical outer partitions were used for paired model comparisons where applicable. Imputation, transformation, centering/scaling, and parameter selection were

176 confined to training folds. AUC, Brier score, calibration intercept, and calibration slope were
177 calculated from the mean held-out predictions. Confidence intervals were based on 1,000
178 patient-level bootstrap resamples of the fixed mean predictions and outcomes, without refitting
179 the models; these intervals therefore do not include full model-development uncertainty. No
180 univariable P-value screening or stepwise variable selection was used [20-26]. A reference
181 coefficient specification is provided in the supplementary material; primary estimates reflect
182 repeated internal validation rather than a single bedside equation. For descriptive Table 1
183 comparisons, continuous variables were compared using the Mann-Whitney U test and
184 categorical variables using Pearson's chi-square test (Fisher's exact test for sparse expected
185 counts). These P values were not used for predictor selection or model development.

186 Temporal sensitivity analysis

187 The temporal sensitivity analysis was based on institutionally deidentified
188 hospitalization/coronary-angiography dates associated with the 2020-2026 study-period
189 cohort. These dates were shifted on a patient-specific basis within +/-182 days; a six-month
190 buffer around the shifted-date split was excluded to reduce potential misclassification of
191 temporal ordering. Legacy WBC_test_time fields were not used for temporal allocation. The
192 retained analysis record identifies an earlier development group, an excluded buffer, and a
193 later fixed-evaluation group, but does not retain a source-verified calendar cutpoint. We
194 therefore report the groups by analysis role rather than supply the unsupported calendar cutoff
195 from an earlier draft. The groups were development N=1,001 (AMI=196), buffer N=271
196 (AMI=79), and later N=548 (AMI=178). This temporal sensitivity analysis is not formal temporal
197 validation.

198 Phenotype and missingness sensitivity analyses

199 High-specificity phenotype and strict-control results were treated as sensitivity analyses. The
200 high-specificity comparison comprised 138 of the 178 later-group AMI records plus 288 strict

CAD/angina controls; the other 40 later-group AMI records did not meet the prespecified diagnosis-text rule and should not be interpreted as adjudicated misclassifications. Fibrinogen was missing in 115/1,820 patients (6.32%), including 27/453 AMI patients (5.96%) and 88/1,367 non-AMI CAD patients (6.44%). Core-7 and Enhanced complete-case models were compared in the same 1,705 patients and identical validation partitions; no fibrinogen imputation was used in this comparison. Fibrinogen was not used to re-adjudicate all AMI phenotypes. PDW was evaluated separately and was not retained because its re-entry criteria were not satisfied.

Reporting and reproducibility

An exploratory decision-curve analysis used the same fibrinogen-complete N=1,705 patients and held-out predictions for the four clinical models, with treat-all and treat-none references over thresholds 0.05-0.50 [27]; it does not establish clinical utility or treatment benefit.

Reporting followed TRIPOD+AI 2024, with PROBAST+AI 2025 used for structured self-audit and STROBE retained for the observational design [28-30]. Figures were generated in R from validated analysis outputs with separate source-data files and scripts. Detailed predictor definitions, preprocessing specifications, model stability results, and sensitivity analyses are provided in Additional files 1 and 2.

Use of generative AI tools

Generative AI tools were used to assist with English-language polishing and code drafting under author supervision. All analysis code was reviewed and tested by the authors, and all data handling, statistical analyses, numerical results, references, interpretations, and final manuscript content were verified by the human authors, who retain full responsibility for the work. No generative AI was used to create patient data or scientific figures.

224 Results

225 Study population

226 Of the 2,279 unique patients, 1,944 had a discharge diagnosis: 456 definite AMI, 1,372 definite
227 non-AMI CAD, five ambiguous MI mentions, and 111 outside the CAD/AMI target. The
228 remaining 335 patients had no discharge diagnosis. Among 1,828 patients with a definite AMI
229 or non-AMI CAD phenotype, eight lacked one or more required core CBC measurements
230 (three AMI and five non-AMI CAD). The final primary cohort included 1,820 patients (453 AMI
231 and 1,367 non-AMI CAD). The strict-control sensitivity cohort included 1,517 patients (453 AMI
232 and 1,064 controls). Figure 1 summarizes source-record consolidation and cohort selection.

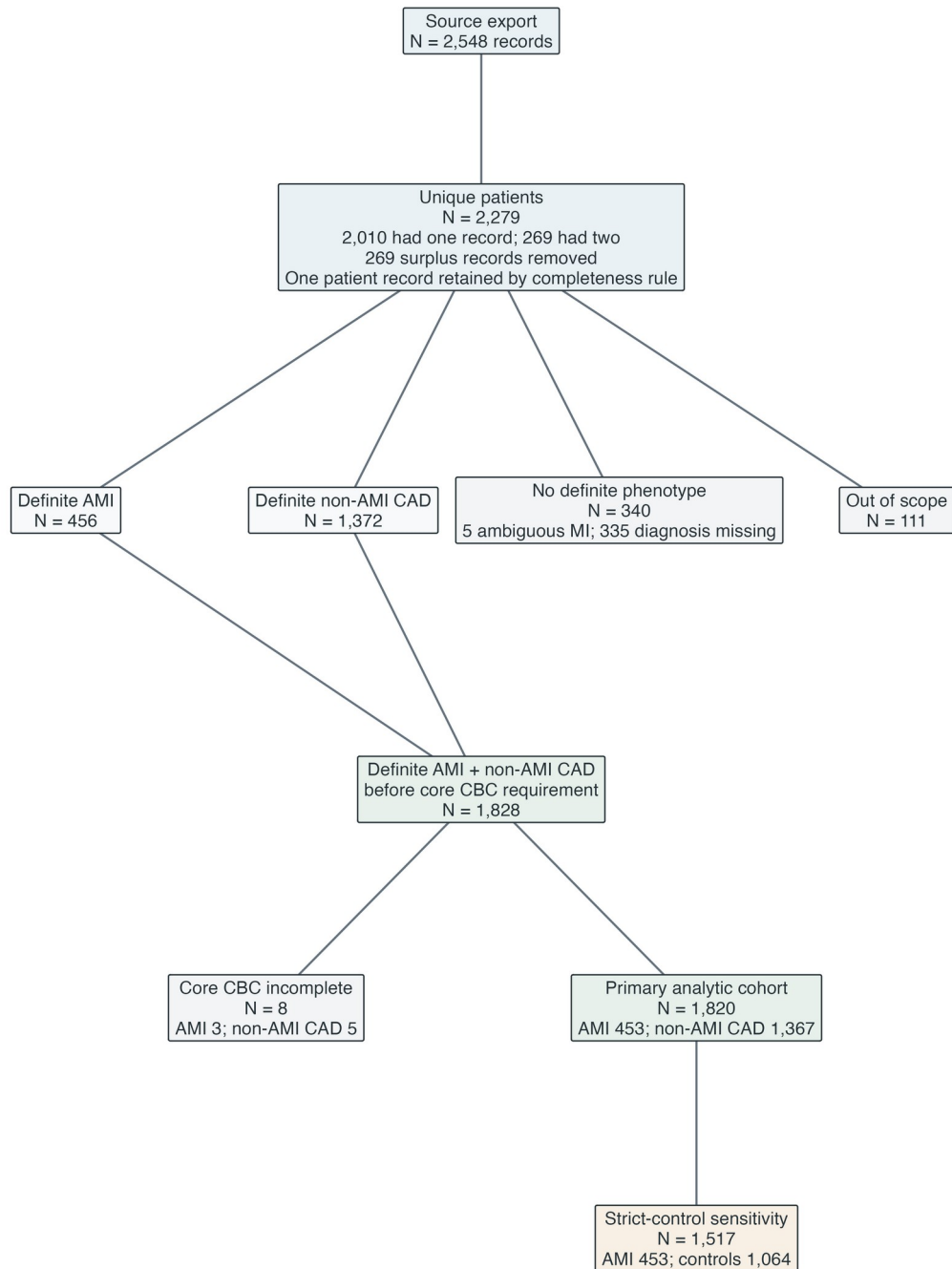


Figure 1. Study population flow. Source-record consolidation, discharge-diagnosis classification, and formation of the primary and strict-control cohorts. Source records are not assumed to represent distinct admissions.

Table 1. Baseline characteristics of the primary cohort according to AMI phenotype

Variable	Unit	Overall (N=1820)	Definite AMI (N=453)	Definite non-AMI CAD (N=1367)	P-value
Age	years	58.8 [50.4, 67.7]	57.4 [48.9, 66.6]	59.3 [50.9, 68.2]	0.004

Variable	Unit	Overall (N=1820)	Definite AMI (N=453)	Definite non-AMI CAD (N=1367)	P-value
Male sex	n/N (%)	1452/1820 (79.8%)	376/453 (83.0%)	1076/1367 (78.7%)	0.049
Hypertension	n/N (%)	927/1741 (53.2%)	201/435 (46.2%)	726/1306 (55.6%)	<0.001
Diabetes	n/N (%)	459/1772 (25.9%)	109/442 (24.7%)	350/1330 (26.3%)	0.491
Neutrophils	10 ⁹ /L	4.4 [3.5, 6.0]	5.7 [4.0, 8.5]	4.2 [3.3, 5.3]	<0.001
Lymphocytes	10 ⁹ /L	2.0 [1.5, 2.5]	1.8 [1.3, 2.4]	2.0 [1.6, 2.5]	<0.001
Monocytes	10 ⁹ /L	0.5 [0.4, 0.7]	0.6 [0.4, 0.8]	0.5 [0.4, 0.6]	<0.001
Platelets	10 ⁹ /L	215.0 [178.0, 255.0]	224.0 [183.0, 266.0]	212.0 [175.5, 253.0]	<0.001
MPV	fL	10.3 [9.5, 11.1]	10.3 [9.6, 11.1]	10.3 [9.5, 11.1]	0.673
RDW	%	13.1 [12.6, 13.6]	12.9 [12.4, 13.4]	13.1 [12.7, 13.6]	<0.001
Hemoglobin	g/L	142.0 [132.0, 152.0]	144.0 [134.0, 155.0]	141.0 [131.0, 152.0]	0.009
Fibrinogen	g/L	3.5 [3.1, 4.1]	3.8 [3.2, 4.5]	3.5 [3.1, 3.9]	<0.001

Continuous variables are presented as median [interquartile range] unless otherwise stated. P-values compare definite AMI with definite non-AMI CAD and were calculated using the Mann-Whitney U test for continuous variables and Pearson's chi-square test for categorical variables; they are descriptive and were not used for model development. Available-case denominators were age 1820 (AMI 453; non-AMI CAD 1367), hypertension 1741, diabetes 1772, and fibrinogen 1705.

Hematologic model

Core-7 combined immune-cell, platelet-related, and erythrocyte-related measurements, while fibrinogen added a secondary coagulation component. These groupings were used to organize the clinical interpretation of the model and were not treated as independent causal pathways.

Internal model performance

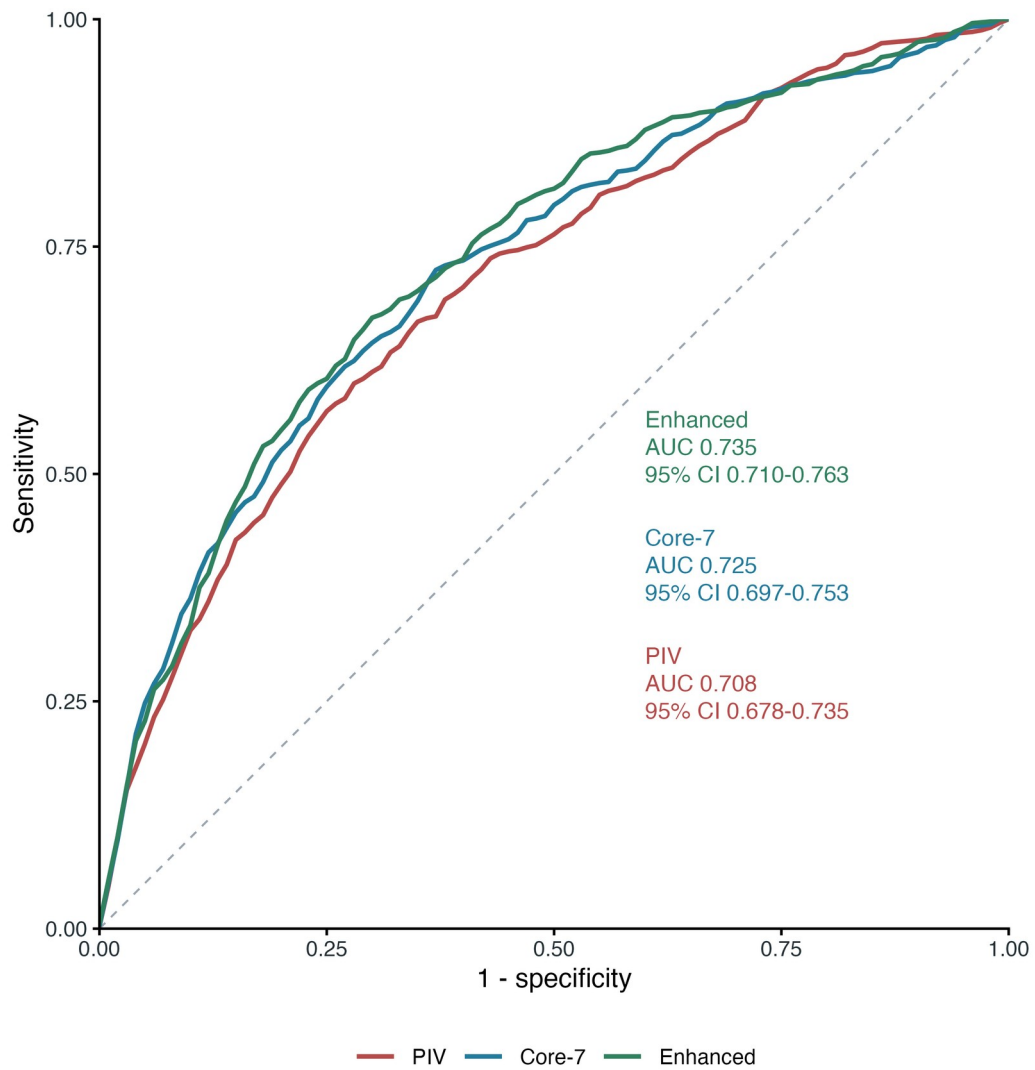
Among the prespecified traditional inflammatory indices, PIV had the highest discriminative ability (AUC 0.708, 95% CI 0.678-0.735). Core-7 yielded an AUC of 0.725 (95% CI 0.697-0.753), corresponding to a modest improvement over PIV (Δ AUC 0.017, 95% CI 0.003-0.031). The Enhanced model achieved an AUC of 0.735 (95% CI 0.710-0.763) and improved on Core-7 by 0.010 (95% CI 0.003-0.017). Calibration intercept and slope were -0.026 and 0.974 for Core-7 and -0.022 and 0.978 for Enhanced. Calibration plots used the same patient-level mean held-out predictions as these statistics; calibration-group sizes, AMI counts, and

predicted-probability summaries are reported in Additional file 2, Table S6. Results are summarized in Table 2 and Figures 2 and 3.

Table 2. Internal cross-validated model performance and paired comparisons

Model / comparison	N	AMI	AUC (95% CI)	Δ AUC (95% CI)	Brier	Calibration intercept	Calibration slope
PIV	1820	453	0.708 (0.678–0.735)	—	0.1667	–0.008	0.992
Core-7	1820	453	0.725 (0.697–0.753)	—	0.1628	–0.026	0.974
Enhanced	1820	453	0.735 (0.710–0.763)	—	0.1619	–0.022	0.978
Core-7 minus PIV	1820	—	—	0.017 (0.003–0.031)	—	—	—
Enhanced minus Core-7	1820	—	—	0.010 (0.003–0.017)	—	—	—

PIV, pan-immune-inflammation value. Enhanced = Core-7 plus fibrinogen. Performance estimates are based on repeated internal cross-validation.



259

260 Figure 2. Primary model discrimination. Cross-validated receiver operating characteristic curves for PIV, Core-7,
261 and the fibrinogen-enhanced model.

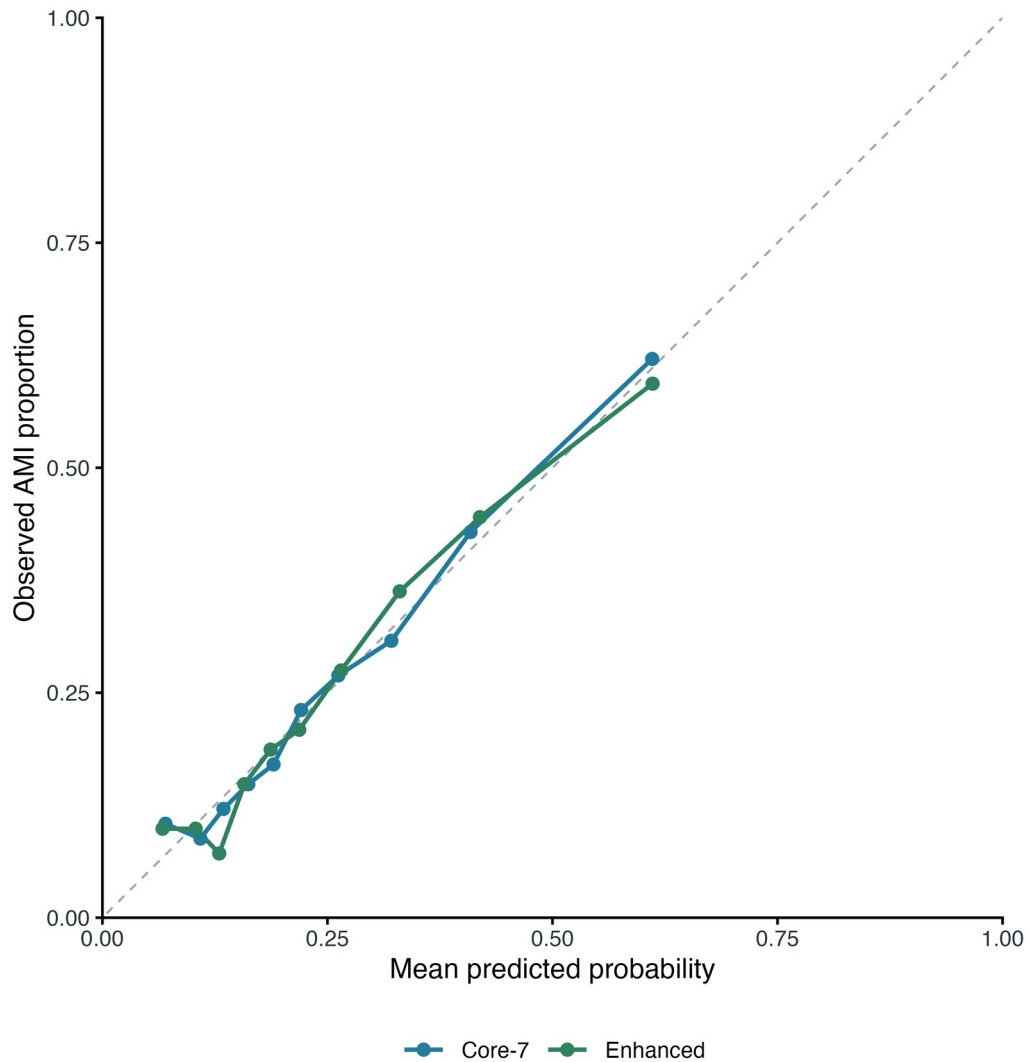


Figure 3. Model calibration. Observed AMI proportions plotted against mean cross-validated predicted probabilities for Core-7 and the Enhanced model.

Clinical incremental value and fibrinogen enhancement

The prespecified clinical baseline model included age, sex, hypertension, and diabetes; smoking was not available. In the same fibrinogen-complete sample (N=1,705; AMI=426), AUCs were 0.546 for Clinical, 0.717 for Clinical+PIV, 0.730 for Clinical+Core-7, and 0.741 for Clinical+Enhanced (95% CIs, 0.515-0.574, 0.689-0.745, 0.701-0.755, and 0.712-0.770, respectively); corresponding Brier scores were 0.187, 0.166, 0.162, and 0.161. Clinical covariate missingness in the primary cohort was 79/1,820 (4.34%) for hypertension and

48/1,820 (2.64%) for diabetes; both were imputed using training-fold medians, while age and sex were complete. On identical patients and outer folds, Clinical+Core-7 versus Clinical+PIV Δ AUC was 0.0130 (95% CI 0.0004-0.0251). Additional file 2, Table S8 reports the complete paired comparisons and calibration results.

Temporal and diagnostic sensitivity analyses

The temporal sensitivity analysis used the deidentified hospitalization/coronary-angiography date axis from the 2020-2026 study period with a six-month buffer. The exact calendar cutpoint was not retained in the available analysis record, so the unsupported cutoff from an earlier draft is not reported. The groups were development N=1,001 (AMI=196), buffer N=271 (AMI=79), and later N=548 (AMI=178); the earlier group was used for development, the later group for fixed evaluation, and the buffer was excluded. In the later group, Core-7, PIV, and Enhanced AUCs were 0.701, 0.699, and 0.719, respectively. Core-7 minus PIV Δ AUC was 0.002 (95% CI -0.021 to 0.025), with an interval including zero. The incremental advantage was attenuated in this temporal sensitivity analysis, which is not formal temporal validation. The high-specificity comparison contained 138 strict-text AMI cases and 288 strict CAD/angina controls (N=426); this discharge-text rule was not independent clinical adjudication. Exploratory decision-curve results for the four clinical models are shown in Supplementary Figure S1 and do not establish clinical utility.

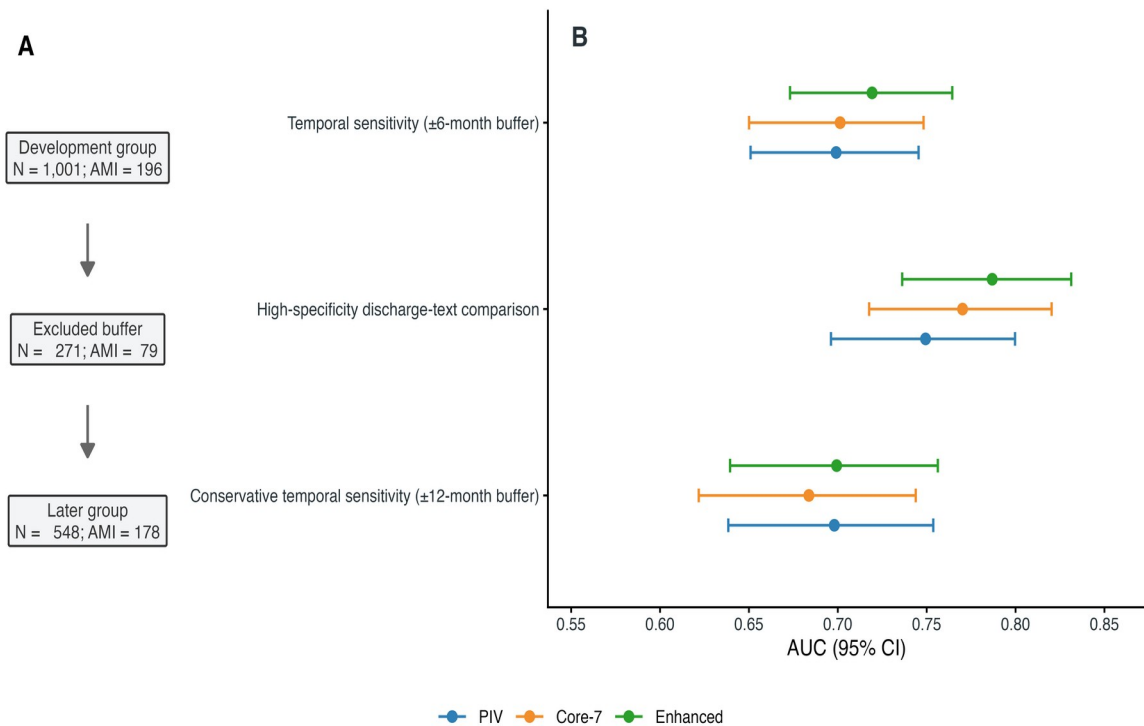


Figure 4. Temporal sensitivity analysis. (A) Development group, excluded buffer, and later group. (B) AUCs with 95% confidence intervals across the later-group, high-specificity discharge-text, and conservative temporal sensitivity comparisons.

Table 3. Robustness analyses across diagnostic and temporal sensitivity cohorts

Analysis	N	AMI	Controls	PIV AUC	Core-7 AUC	Enhanced AUC
Temporal sensitivity, later group	548	178	370	0.699 (0.651–0.745)	0.701 (0.650–0.748)	0.719 (0.673–0.764)
High-specificity discharge-text phenotype comparison	426	138	288	0.749 (0.696–0.800)	0.770 (0.718–0.820)	0.787 (0.736–0.831)
Conservative temporal sensitivity (±12-month buffer)	361	110	251	0.698 (0.638–0.754)	0.684 (0.622–0.744)	0.699 (0.639–0.756)
Alternative date-shift sensitivity cohort	602	198	404	0.704	0.707	0.725
Strict-control diagnostic cohort	1517	453	1064	0.730 (0.701–0.757)	0.747 (0.718–0.775)	0.759 (0.733–0.785)

For the later group in the temporal sensitivity analysis, Core-7 minus PIV Δ AUC was 0.002 (95% CI -0.021 to 0.025), and the interval included zero. This analysis is not formal temporal validation. Alternative date-shift estimates are presented as point estimates because confidence intervals were not available from those analyses.

Predictor stability

In the primary stability analysis, neutrophils, lymphocytes, monocytes, platelets, and RDW were selected in all resampling iterations, MPV in 99%, and hemoglobin in 95%. These results indicate that information from leukocyte, platelet, and erythrocyte measurements contributed reproducibly to the multivariable model. Fibrinogen was retained only in the Enhanced model. PDW was evaluated during model development but excluded because of temporal and measurement instability. Additional stability and domain-level results are provided in Additional file 1 and Additional file 2.

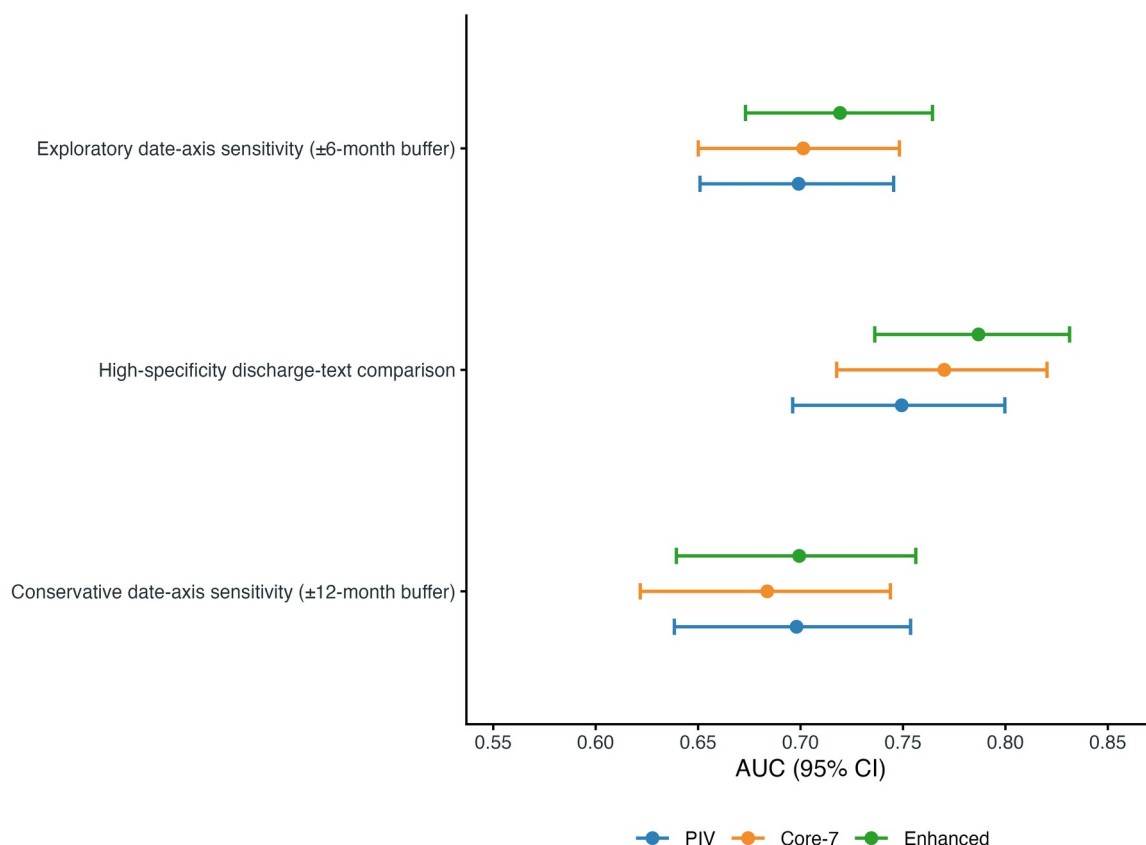


Figure 5. Robustness across sensitivity analyses. AUC estimates and available 95% confidence intervals for PIV, Core-7, and Enhanced across the sensitivity comparisons. The alternative date-shift point estimates without confidence intervals are not plotted.

Discussion

In this CAD-restricted cohort, combining routinely measured hematologic variables provided moderate discrimination of the AMI phenotype. Core-7 achieved an AUC of 0.725 (95% CI 0.697-0.753), with a statistically significant but modest improvement over PIV (Δ AUC 0.017, 95% CI 0.003-0.031). The clinical relevance of this finding lies less in the absolute size of the AUC increase than in what it suggests about routine laboratory information: cell counts, platelet size, and erythrocyte indices retain complementary information when evaluated together rather than being reduced to a single predefined ratio [3-17].

The biological rationale for this multidimensional approach is consistent with the pathophysiology of acute coronary events. Neutrophilia and lymphocyte depletion accompany inflammatory and stress responses, while circulating monocytes participate in plaque inflammation and the immune response after myocardial injury [1,3-12]. Platelet count and MPV may reflect complementary aspects of platelet activation, consumption, and renewed platelet production during acute coronary thrombosis [13,14]. RDW and hemoglobin can be influenced by inflammation, erythropoietic stress, comorbidity, and oxygen-carrying reserve [15,16]. Fibrinogen, as both an acute-phase protein and a coagulation substrate, adds information from the thrombo-inflammatory axis that is not directly captured by the CBC [17]. These mechanisms provide a clinical rationale for considering the variables jointly, while the individual regression coefficients should not be interpreted as causal effects or as evidence that any single marker is protective or harmful.

The incremental advantage of Core-7 over PIV was attenuated in the temporal sensitivity analysis. Core-7 and PIV had AUCs of approximately 0.701 and 0.699, respectively, and the paired confidence interval included zero. The temporal sensitivity analysis used deidentified

334 hospitalization/coronary-angiography dates from the study period but is not formal temporal
335 validation. Independent cohorts with standardized clinical time points will be needed to
336 determine how well this performance generalizes.

337 Fibrinogen provided a modest additional increment over Core-7 in the primary internal
338 analysis; an increment was also observed in the temporal sensitivity analysis, which was not
339 formal temporal validation. This finding is clinically plausible because fibrinogen reflects both
340 acute-phase activation and the coagulation milieu [17]. However, fibrinogen was not available
341 as uniformly as the CBC variables, and the Enhanced model was not consistently superior to
342 PIV across all analyses. The inverse RDW coefficient and changes in the direction of MPV,
343 hemoglobin, and platelet associations between crude and multivariable analyses also caution
344 against assigning simple mechanistic interpretations to individual coefficients [13-16]. The
345 value of the model lies in the combined hematologic pattern rather than in redefining these
346 measurements as isolated AMI risk factors.

347 Because all seven Core-7 variables are part of routine complete blood count testing, the model
348 could in principle be calculated automatically within a laboratory information system or
349 electronic health record without requiring an additional assay. Such integration may eventually
350 provide background hematologic context during the early assessment of patients with
351 suspected acute coronary disease. The present study did not test real-time triage, treatment
352 decisions, or performance before cardiac biomarker results became available. No standardized
353 troponin comparator was available because assay platform, reference limits, serial change,
354 and index timing were not sufficiently harmonized. The findings should therefore be interpreted
355 as internal phenotype-discrimination results, not as evidence of diagnostic or triage readiness.

356 Several features strengthen the analysis, including predefined diagnostic mapping,
357 preprocessing restricted to the training data during cross-validation, common cross-validated
358 predictions for model comparisons, assessment of predictor stability, and a temporal sensitivity
359 analysis. The study also has important limitations. It is retrospective and single-center; AMI

status was based primarily on discharge-diagnosis text rather than independent adjudication with serial biomarkers, electrocardiography, imaging, and full clinical context; and no independent external cohort was available. The source export lacked a complete eligible-patient screening log, so consecutive enrollment could not be confirmed. The extract did not contain validated index-relative prior MI or prior revascularization fields, so their prevalence and predictor roles could not be assessed. Discharge-diagnosis availability contributed to source-row ranking, and the effect of this availability-based selection cannot be quantified without the upstream extraction logic. Because admission dates were unavailable for many patients, age was derived using the available CBC/WBC timestamp as a fallback reference date and was therefore not uniformly anchored to the index angiography hospitalization. Although the analyzed hematologic measurements were associated with the target hospitalization, exact sampling times relative to admission, coronary angiography, and AMI diagnosis could not be reconstructed uniformly from the exported research dataset. Fibrinogen measurements were less complete than CBC measurements. The temporal sensitivity analysis is not formal temporal validation. These limitations define the next step: independent evaluation with a verified clinical index time, harmonized laboratory measurements, adjudicated AMI status, and prospective assessment of calibration and clinical utility [20,21,26-30].

Conclusions

Multidimensional analysis of routine CBC variables provided modest additional information beyond a fixed PIV ratio in the internal evaluation. Core-7 retained moderate discrimination in the temporal sensitivity analysis, although its incremental advantage over PIV was attenuated. Addition of fibrinogen provided a further small increment. These findings warrant independent evaluation with prospectively time-anchored laboratory measurements and standardized clinical adjudication; they do not establish diagnostic or triage readiness.

List of abbreviations

AMI, acute myocardial infarction; AUC, area under the receiver operating characteristic curve; CAD, coronary artery disease; CBC, complete blood count; CI, confidence interval; CK-MB, creatine kinase-MB; HRR, hemoglobin-to-red cell distribution width ratio; MLR, monocyte-to-lymphocyte ratio; MPV, mean platelet volume; NLR, neutrophil-to-lymphocyte ratio; NSTEMI, non-ST-segment elevation myocardial infarction; PCI, percutaneous coronary intervention; PDW, platelet distribution width; PIV, pan-immune-inflammation value; PLR, platelet-to-lymphocyte ratio; RDW, red cell distribution width; ROC, receiver operating characteristic; SII, systemic immune-inflammation index; SIRI, systemic inflammation response index; STEMI, ST-segment elevation myocardial infarction.

Declarations

Clinical trial number

Clinical trial number: not applicable.

Ethics approval

The study was approved by the Ethics Committee of Xinjiang Medical University (approval no. K202602-10). All methods were performed in accordance with relevant guidelines and regulations.

Consent to participate

Given the retrospective nature of the study, the requirement for informed consent was waived by the Ethics Committee.

Consent for publication

Not applicable.

Availability of data and materials

The clinical dataset is not publicly available because of institutional, ethical, and participant-privacy restrictions. Deidentified data may be available from the corresponding authors on reasonable request and subject to applicable institutional and ethics approval. Non-identifying analysis and figure-generation code is available from the corresponding authors on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

JA: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization, Project administration. MP: Methodology, Validation, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. QX: Investigation, Resources, Data curation, Validation, Writing - review & editing. QP: Investigation, Resources, Data curation, Validation, Writing - review & editing. WJ: Investigation, Resources, Data curation, Validation, Writing - review & editing. BF: Investigation, Resources, Data curation, Validation, Writing - review & editing. FL: Methodology, Formal analysis, Validation, Writing - review & editing. LZ: Resources,

Validation, Writing - review & editing, Supervision. XL: Conceptualization, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition. YY: Conceptualization, Resources, Validation, Writing - review & editing, Supervision. All authors read and approved the final manuscript.

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Additional files

Additional file 1. DOCX. Supplementary methods and figures. Contains supplementary methodological details and Figures S1-S4.

Additional file 2. XLSX. Supplementary tables. Contains Supplementary Tables S1-S13 covering variable definitions, phenotype mapping, missingness, conventional index formulas, preprocessing, resampling, predictor stability, clinical incremental models, fibrinogen complete-case analysis, high-specificity AMI analysis, temporal sensitivity analyses, the exploratory simplified score, and the reference model specification.

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